

ubuntu@ip-172-31-10-181:~\$ #Lab56. Basic Cron Log Inspection_lab

ubuntu@ip-172-31-10-181:~\$ #OBJECTIVES: Understand the importance of cron logs in monitoring scheduled tasks. Learn how to locate and interpret cron job entries in system log files. Gain skills to search and filter specific log entries related to cron jobs.

ubuntu@ip-172-31-10-181:~\$ #Task1: Locate_the_log_files

ubuntu@ip-172-31-10-181:~\$

ubuntu@ip-172-31-10-181:~\$ ls -l /var/log

total 4172

lrwxrwxrwx	1 root	root	39 Oct 22 2025	README ->
../usr/share/doc/systemd/README.logs				
-rw-r--r--	1 root	root	0 Jan 5 07:14	alternatives.log
-rw-r--r--	1 root	root	781 Jan 2 11:17	alternatives.log.1
drwx-----	3 root	root	4096 Dec 20 07:05	amazon
-rw-r-----	1 root	adm	0 Dec 8 07:07	apport.log
drwxr-xr-x	2 root	root	4096 May 29 18:47	apt
-rw-r-----	1 syslog	adm	3073 May 29 18:50	auth.log
-rw-r-----	1 syslog	adm	6182 May 29 18:47	auth.log.1
-rw-r-----	1 syslog	adm	6441 Jan 24 06:06	auth.log.2.gz
-rw-r-----	1 syslog	adm	6701 Jan 5 07:14	auth.log.3.gz
-rw-r-----	1 syslog	adm	1953 Dec 27 13:22	auth.log.4.gz
-rw-rw----	1 root	utmp	0 May 29 18:47	btmptmp
-rw-rw----	1 root	utmp	3840 Jan 10 10:11	btmptmp.1
drwxr-xr-x	2 _chrony	_chrony	4096 Dec 3 13:16	chrony
-rw-r-----	1 root	adm	49747 May 29 18:48	cloud-init-output.log
-rw-r-----	1 syslog	adm	1065233 May 29 18:48	cloud-init.log
-rw-r-----	1 syslog	adm	1108866 Jan 6 05:36	cloud-init.log.1
drwxr-xr-x	2 root	root	4096 Dec 20 07:00	cups
drwxr-xr-x	2 cups-browsed	lpadmin	4096 Dec 3 13:25	cups-browsed
drwxrwsrwt	2 dcvt	dcvt	4096 May 29 18:48	dcvt
-rw-r--r--	1 root	root	51 Jan 2 11:00	dcvt-setup.log
drwxr-xr-x	2 root	root	4096 Jul 25 2025	dist-upgrade
-rw-r-----	1 root	adm	50205 May 29 18:47	dmesg
-rw-r-----	1 root	adm	48255 Jan 24 06:07	dmesg.0
-rw-r-----	1 root	adm	14921 Jan 10 13:23	dmesg.1.gz
-rw-r-----	1 root	adm	14968 Jan 10 10:00	dmesg.2.gz
-rw-r-----	1 root	adm	14765 Jan 7 09:56	dmesg.3.gz
-rw-r-----	1 root	adm	14981 Jan 7 06:42	dmesg.4.gz
-rw-r--r--	1 root	root	0 May 29 18:47	dpkg.log
-rw-r--r--	1 root	root	80215 Jan 2 11:19	dpkg.log.1
-rw-r--r--	1 root	root	370 Dec 20 07:05	dpkg.log.2.gz
-rw-r--r--	1 root	root	0 Dec 8 07:07	fontconfig.log
drwx-x-x	2 root	gdm	4096 Dec 3 13:30	gdm3
-rw-r--r--	1 root	root	1435 May 29 18:47	gpu-manager.log
drwxr-xr-x	3 root	root	4096 Dec 3 13:22	hp
drwxr-sr-x+	3 root	systemd-journal	4096 Dec 3 13:16	journal
-rw-r-----	1 syslog	adm	89 May 29 18:48	kern.log

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-rw-r----- 1 syslog      adm      63460 May 29 18:47 kern.log.1
-rw-r----- 1 syslog      adm     100215 Jan 24 06:06 kern.log.2.gz
-rw-r----- 1 syslog      adm     64478 Jan  5 07:14 kern.log.3.gz
-rw-r----- 1 syslog      adm     13015 Dec 27 13:22 kern.log.4.gz
drwxr-xr-x  2 landscape   landscape 4096 Dec  3 13:17 landscape
-rw-rw-r--  1 root        utmp      292292 May 29 18:50 lastlog
drwxr-xr-x  2 root        root        4096 Nov 24  2025 openvpn
drwx----- 2 root        root        4096 Dec  3 13:16 private
drwxr-xr-x  2 root        root        4096 May 29 18:47 sessions
drwx----- 2 speech-dispatcher root      4096 Apr  7  2024 speech-dispatcher
drwxr-x--  2 root        root        4096 Aug 12  2025 sssd
-rw-r----- 1 syslog      adm     162992 May 29 18:52 syslog
-rw-r----- 1 syslog      adm     316020 May 29 18:47 syslog.1
-rw-r----- 1 syslog      adm     433316 Jan 24 06:06 syslog.2.gz
-rw-r----- 1 syslog      adm     304459 Jan  5 07:14 syslog.3.gz
-rw-r----- 1 syslog      adm     50546 Dec 27 13:22 syslog.4.gz
drwxr-xr-x  2 root        root        4096 Jan 24 06:06 sysstat
-rw-r-r--  1 root        root         0 Dec  8 07:07 ubuntu-advantage-apt-hook.log
drwxr-x--  2 root        adm         4096 Jan  2 10:56 unattended-upgrades
-rw-rw-r--  1 root        utmp      91392 May 29 18:50 wtmp

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ubuntu@ip-172-31-10-181:~\$

ubuntu@ip-172-31-10-181:~\$ less /var/log/syslog

ubuntu@ip-172-31-10-181:~\$

ubuntu@ip-172-31-10-181:~\$ #Task2:Task 2: Search for Specific Cron Job Entries

ubuntu@ip-172-31-10-181:~\$ grep CRON /var/log/syslog

ubuntu@ip-172-31-10-181:~\$ grep cron /var/log/cron

grep: /var/log/cron: No such file or directory

ubuntu@ip-172-31-10-181:~\$ grep cron /var/log/syslog

2026-05-29T18:52:51.619981+00:00 ip-172-31-10-181 anacron[930]: Job `cron.daily' started

2026-05-29T18:52:51.625569+00:00 ip-172-31-10-181 anacron[4999]: Updated timestamp for job `cron.daily' to 2026-05-29

2026-05-29T18:52:51.636150+00:00 ip-172-31-10-181 anacron[930]: Job `cron.daily' terminated

ubuntu@ip-172-31-10-181:~\$ grep 'Jan 15' /var/log/syslog | grep CRON

ubuntu@ip-172-31-10-181:~\$ grep 'May 29' /var/log/syslog | grep CRON

ubuntu@ip-172-31-10-181:~\$ grep 'May 29' /var/log/syslog | grep cron

ubuntu@ip-172-31-10-181:~\$ sudo grep -i CRON /var/log/syslog | tail -n 200

2026-05-29T18:47:55.748314+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1207]: [2026-05-29 18:47:55] Lab: 56. Basic Cron Log Inspection_lab

2026-05-29T18:47:55.772818+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1240]: [2026-05-29 18:47:55] CloudWatch stream:

128539_khanzarbakht_gmail_com_56_Basic_Cron_Log_Inspection_lab_IP_100_24_38_51

2026-05-29T18:48:11.094091+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1882]: [2026-05-29 18:48:11] Stream:

128539_khanzarbakht_gmail_com_56_Basic_Cron_Log_Inspection_lab_IP_100_24_38_51

2026-05-29T18:52:51.619981+00:00 ip-172-31-10-181 anacron[930]: Job `cron.daily' started

2026-05-29T18:52:51.625569+00:00 ip-172-31-10-181 anacron[4999]: Updated timestamp for job `cron.daily' to 2026-05-29

```

2026-05-29T18:52:51.636150+00:00 ip-172-31-10-181 anacron[930]: Job `cron.daily' terminated
2026-05-29T18:55:01.714193+00:00 ip-172-31-10-181 CRON[6875]: (root) CMD (command -v debian-
sa1 > /dev/null && debian-sa1 1 1)
ubuntu@ip-172-31-10-181:~$ ^C
ubuntu@ip-172-31-10-181:~$
ubuntu@ip-172-31-10-181:~$ #ANALYSIS:
ubuntu@ip-172-31-10-181:~$ #Concise Report: Cron Log Analysis Source: sudo grep -i CRON
/var/log/syslog | tail -n 200 System: AWS EC2 instance (ip-172-31-10-181) Date: 2026-05-29 Key
Findings: Lab Environment Activity (18:47:55 – 18:48:11) CloudWatch AMI re-initialization script
executed for Lab 56: Basic Cron Log Inspection. Anacron Daily Job (18:52:51) cron.daily job started
and completed successfully. Timestamp updated for 2026-05-29. Regular Cron Job (18:55:01) Root
executed: command -v debian-sa1 > /dev/null && debian-sa1 1 1 → This is the standard sysstat
performance data collection job (runs every 10 minutes). Summary: The logs show normal
cron/anacron activity on a training lab instance. The only active scheduled task visible is the
sysstat (sa1) collector, along with the daily maintenance job via anacron. No errors or failed cron
jobs were observed. Status: Normal □
ubuntu@ip-172-31-10-181:~$
ubuntu@ip-172-31-10-181:~$ #Task3:TROUBLESHOOT_AND_OPTIMIZE_CRON_JOBS
ubuntu@ip-172-31-10-181:~$ grep CRON /var/log/syslog | grep -i 'error\|fail'
awk: warning: escape sequence `\" treated as plain `\"
ubuntu@ip-172-31-10-181:~$ grep CRON /var/log/syslog | grep -i 'error|fail'
ubuntu@ip-172-31-10-181:~$ grep CRON /var/log/syslog | grep -i 'error\fail'
ubuntu@ip-172-31-10-181:~$ grep cron /var/log/syslog | grep -i 'error\fail'
ubuntu@ip-172-31-10-181:~$
ubuntu@ip-172-31-10-181:~$ grep -iE 'CRON|anacron' /var/log/syslog | grep -iE 'error|fail|failed|
failure'
ubuntu@ip-172-31-10-181:~$ grep -iE 'cron|anacron' /var/log/syslog | grep -iE 'error|fail|failed|
failure'
ubuntu@ip-172-31-10-181:~$ sudo grep -iE 'CRON|anacron' /var/log/syslog | tail -n 200
2026-05-29T18:47:55.748314+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1207]: [2026-05-29
18:47:55] Lab: 56. Basic Cron Log Inspection_lab
2026-05-29T18:47:55.772818+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1240]: [2026-05-29
18:47:55] CloudWatch stream:
128539_khanzarbakht_gmail_com_56_Basic_Cron_Log_Inspection_lab_IP_100_24_38_51
2026-05-29T18:48:11.094091+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1882]: [2026-05-29
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sa1 > /dev/null && debian-sa1 1 1)
2026-05-29T18:57:51.620878+00:00 ip-172-31-10-181 anacron[930]: Job `cron.weekly' started
2026-05-29T18:57:51.626092+00:00 ip-172-31-10-181 anacron[9200]: Updated timestamp for job
`cron.weekly' to 2026-05-29
2026-05-29T18:57:51.626995+00:00 ip-172-31-10-181 anacron[930]: Job `cron.weekly' terminated

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2026-05-29T19:02:51.621288+00:00 ip-172-31-10-181 anacron[930]: Job `cron.monthly' started
2026-05-29T19:02:51.626484+00:00 ip-172-31-10-181 anacron[930]: Job `cron.monthly' terminated
2026-05-29T19:02:51.627188+00:00 ip-172-31-10-181 anacron[13078]: Updated timestamp for job
`cron.monthly' to 2026-05-29
2026-05-29T19:02:51.627466+00:00 ip-172-31-10-181 systemd[1]: anacron.service: Deactivated
successfully.
2026-05-29T19:02:51.627549+00:00 ip-172-31-10-181 anacron[930]: Normal exit (3 jobs run)
2026-05-29T19:05:01.749561+00:00 ip-172-31-10-181 CRON[14941]: (root) CMD (command -v
debian-sa1 > /dev/null && debian-sa1 1 1)
ubuntu@ip-172-31-10-181:~\$ ^C
ubuntu@ip-172-31-10-181:~\$ #No_adjustments_currently_needed
ubuntu@ip-172-31-10-181:~\$ #LabCompleted
ubuntu@ip-172-31-10-181:~\$ sudo grep -iE 'CRON|anacron' /var/log/syslog | tail -n 200
2026-05-29T18:47:55.748314+00:00 ip-172-31-10-181 cloudwatch-ami-reinit.sh[1207]: [2026-05-29
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`cron.weekly' to 2026-05-29
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2026-05-29T19:02:51.621288+00:00 ip-172-31-10-181 anacron[930]: Job `cron.monthly' started
2026-05-29T19:02:51.626484+00:00 ip-172-31-10-181 anacron[930]: Job `cron.monthly' terminated
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debian-sa1 > /dev/null && debian-sa1 1 1)
ubuntu@ip-172-31-10-181:~\$ #LabDone
ubuntu@ip-172-31-10-181:~\$